

Generation Interconnection System Impact Study Report

For Queue Project Q16-0690

Allen 100 kV
October 02, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0690. The Interconnection Customer proposes a 164.1 MW solar photovoltaic facility interconnecting at the Allen 100 kV substation in the Indiana area. The request was received on October 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 337 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,619,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hillcrest to Wayne Trace 100kV
Contingency: Wyne Trace to Darlington Road 100kV
Loading: 147.6% (Rating: 153 MVA)
DFAX: 50.8%
Pre-project loading: 93.1%
2. Facility: Aep Pettit Avenue to Allen 100kV
Contingency: Wayne Trace to Allen 100kV
Loading: 142.0% (Rating: 298 MVA)
DFAX: 100.0%
Pre-project loading: 86.9%
3. Facility: Wayne Trace to Allen 100kV
Contingency: Allen to Aep Pettit Avenue 100kV
Loading: 142.0% (Rating: 298 MVA)
DFAX: 100.0%
Pre-project loading: 86.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hillcrest to Wayne Trace 100kV	Rebuild line with higher-capacity conductor	\$1,266,000	38% (\$475,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aep Pettit Avenue to Allen 100kV	Rebuild line with higher-capacity conductor	\$8,049,000	38% (\$3,020,000)
Wayne Trace to Allen 100kV	Rebuild line with higher-capacity conductor	\$5,662,000	38% (\$2,124,000)
	Total allocated to project		\$5,619,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.